

Find and Fix: Degree–Radian and Quadrant–Sign Errors

Answer Key

Precalculus

Quick Answers

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|---|---|
| 1. (a) correct degree measure = 150 degrees, — | degrees, — |
| 2. (a) correct exact value = $\frac{(-1)\sqrt{3}}{2}$, — | 5. (a) correct exact value = $-\frac{1}{2}$, — |
| 3. (a) correct solutions = 30, 150 degrees, — | 6. (a) correct value in radian mode = 0.9093, — |
| 4. (a) all solutions in the interval = 135, 315 | 7. (a) exact value of cosine = $\frac{4}{5}$, — |
| | 8. (a) exact solutions in radians = $\frac{2\pi}{3}$, $\frac{4\pi}{3}$, — |
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What is verified

8 of 16 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-TF.A.2 – problems 1, 2, 5, 6, 7

HSF-TF.B.5 – problems 3, 4, 8

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

1. If they got 0.0457: multiplied by pi over 180 instead of 180 over pi, converting in the wrong direction
 2. If they got 0.866: kept the reference angle's positive value instead of applying the sign for the third quadrant
 4. If they got -45: reported the inverse tangent output unchanged, so the angle fell outside the interval and the second solution was lost
 5. If they got -0.866: attached the correct third quadrant sign but read the cosine of the reference angle instead of its sine
 6. If they got 0.0349: left the calculator in degree mode, so the sine of 2 degrees was returned instead
 7. If they got -0.8: made cosine negative as well, which is the third quadrant rather than the one the given information forces
 8. If they got 120, 240: answered in degrees although the interval was written in radians, so both values sit far outside it
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1. Convert $\frac{5\pi}{6}$ radians to degrees. Student wrote $\frac{5\pi}{6} \cdot \frac{\pi}{180} = 0.0457$.

(a) A half turn is π radians and also 180° , so the two conversion factors are $\frac{\pi}{180}$ (degrees $\rightarrow$ radians) and $\frac{180}{\pi}$ (radians $\rightarrow$ degrees). The measure given is already in radians, so use the second:

$$\frac{5\pi}{6} \cdot \frac{180}{\pi} = \frac{5 \cdot 180}{6} = 150.$$

The π cancels, which is the signal that the direction is right.

150°

(b) *Instructor-judged.* The response must say the factor was used upside down: $\frac{\pi}{180}$ converts degrees into radians, and the given measure was already in radians, so $\frac{180}{\pi}$ was needed. Full credit requires naming the direction; “the answer is too small” alone is partial. A useful check to point out: the π did not cancel, and a leftover π^2 is always a sign the factor was inverted.

2. Evaluate $\cos 210^\circ$. Student wrote $+\frac{\sqrt{3}}{2}$, from reference angle 30° .

(a) $210^\circ = 180^\circ + 30^\circ$, so the reference angle is 30° and the terminal side lies in quadrant III. In quadrant III cosine is negative, so

$$\cos 210^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2} \approx -0.866.$$

$-\frac{\sqrt{3}}{2}$

(b) *Instructor-judged.* The response must locate the fault in the *sign* step, not the reference angle: both the reference angle 30° and the value $\frac{\sqrt{3}}{2}$ are correct, but 210° is in quadrant III, where cosine is negative, so the minus sign has to be attached at the end. Full credit requires naming the quadrant and the sign of cosine there.

3. Solve $2 \sin \theta - 1 = 0$ on $0^\circ \leq \theta < 360^\circ$. Student reported 0.5236 and 2.6180.

(a) $2 \sin \theta = 1$, so $\sin \theta = \frac{1}{2}$. Sine is positive in quadrants I and II, and the reference angle with sine $\frac{1}{2}$ is 30° , so

$$\theta = 30^\circ \quad \text{and} \quad \theta = 180^\circ - 30^\circ = 150^\circ.$$

$\theta = 30^\circ, 150^\circ$

(b) *Instructor-judged.* The response must say the calculator was left in radian mode while the interval was stated in degrees. A strong answer notes that $0.5236 \approx \frac{\pi}{6}$ and $2.6180 \approx \frac{5\pi}{6}$ are the *same two angles* measured in radians, so the mathematics was right and only the mode was wrong. “The numbers are too small” alone is partial credit.

4. Solve $\tan \theta = -1$ on $0^\circ \leq \theta < 360^\circ$. Student reported the single solution -45° .

(a) Tangent is negative in quadrants II and IV, and $\tan 45^\circ = 1$, so the reference angle is 45° :

$$\theta = 180^\circ - 45^\circ = 135^\circ \quad (\text{II}), \quad \theta = 360^\circ - 45^\circ = 315^\circ \quad (\text{IV}).$$

Tangent has period 180° , so a 360° interval always holds two solutions.

$\theta = 135^\circ, 315^\circ$

(b) *Instructor-judged.* Two distinct faults must be named. (i) $\tan^{-1}$ returns an angle strictly between -90° and 90° , so its output must be moved into the requested interval — here by adding 180° . (ii) A 360° interval spans two periods of tangent, so there are two solutions, not one. Naming only one fault is half credit.

5. Evaluate $\sin \frac{7\pi}{6}$. Student wrote $-\frac{\sqrt{3}}{2}$.

(a) $\frac{7\pi}{6} = \pi + \frac{\pi}{6}$, so the reference angle is $\frac{\pi}{6}$ and the angle is in quadrant III, where sine is negative. The sine of $\frac{\pi}{6}$ is $\frac{1}{2}$, not $\frac{\sqrt{3}}{2}$:

$$\sin \frac{7\pi}{6} = -\sin \frac{\pi}{6} = -\frac{1}{2}.$$

$-\frac{1}{2}$

(b) *Instructor-judged.* The response must say the sign was right and the ratio was wrong: $\frac{\sqrt{3}}{2}$ is the *cosine* of $\frac{\pi}{6}$, while its sine is $\frac{1}{2}$. The quadrant fixes only the sign, never which ratio to read. Full credit requires naming the sine-for-cosine swap explicitly.

6. Find $\sin(2)$ with 2 in radians. Student's calculator returned 0.0349.

(a) In radian mode, $\sin(2) = 0.9092974\dots$, so to four decimal places.

0.9093

(b) *Instructor-judged.* The response must say the calculator was in degree mode, so it evaluated $\sin 2^\circ$ rather than the sine of an arc of 2 radians. A strong answer adds the reason the gap is so wide: 2 radians is about 114.6° , an angle in quadrant II not far from the maximum of sine, whereas 2° is nearly a zero angle. Checking that $\sin 2^\circ \approx 0.0349$ reproduces the student's number exactly is the confirmation.

7. Given $\sin \theta = -\frac{3}{5}$ and $\cos \theta > 0$. Student put θ in quadrant III and wrote $\cos \theta = -\frac{4}{5}$.

(a) From the Pythagorean identity, $\cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos \theta = \pm \frac{4}{5}$.

The problem states $\cos \theta > 0$, which selects the positive root.

$\cos \theta = \frac{4}{5}$

(b) *Instructor-judged.* The response must place θ in quadrant IV and give the two-sign argument: a negative sine rules out quadrants I and II, a positive cosine rules out quadrants II and III, and only quadrant IV survives both. The error was to decide the quadrant from the sign of sine alone and then to *derive* the sign of cosine from that guess, discarding the second given. Naming the quadrant without the argument is partial credit.

8. Solve $\cos \theta = -\frac{1}{2}$ on $0 \leq \theta < 2\pi$. Student reported 120 and 240.

(a) Cosine is negative in quadrants II and III, and the reference angle with cosine $\frac{1}{2}$ is $\frac{\pi}{3}$:

$$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \quad (\text{II}), \quad \theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3} \quad (\text{III}).$$

$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

(b) *Instructor-judged.* Two faults, and the check that catches them. (i) The interval was written in radians (0 to 2π), so the answers must be radian measures; the student found the right angles but reported them in degrees. (ii) Neither reported value lies in the stated interval at all, since $2\pi \approx 6.28$. The fix is to multiply each degree measure by $\frac{\pi}{180}$, and the check is to confirm every answer actually falls inside the interval before writing it down.